

# MILAD SHIRANI

U.S. Permanent Resident (Green Card)

Postdoctoral Associate · Yale University · New Haven, CT

[milad.shirani@yale.edu](mailto:milad.shirani@yale.edu) ·  ·  ·  · 

## PROFILE

---

Scientific AI researcher and ML engineer with 10+ years developing physics-based simulation and machine learning methods for nonlinear solids and coupled multiphysics systems. I train neural networks—including graph neural networks—to solve coupled PDE systems by minimizing the total potential energy or the FEM weak-form residual directly, with no labeled solution data and Dirichlet conditions imposed exactly. The networks share their discretization (shape functions, quadrature, boundary conditions) with the finite-element solver, so every result is compared node by node with a converged Newton–Raphson reference. I also build differentiable simulators: a matrix-free Newton–Krylov finite-element solver and autodiff-based meshless peridynamics for fracture and delamination. A neural operator I trained predicted unseen cases  $\sim 35\text{--}250\times$  faster per query than FEM on a benchmark problem, and across four coupled systems my networks match converged finite-element solutions to within 0.05–3%. Open-source ML development and startup experience translating scientific ML methods into engineering models.

## TECHNICAL SKILLS

---

|                            |                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scientific AI / ML</b>  | Neural operators (DeepONets, FNOs); physics-informed neural networks (PINNs); graph neural networks (GNNs; message-passing, graph convolution); surrogate modeling; differentiable scientific computing; variational energy-minimizing networks; weak-form (Galerkin) residual training; physics-constrained learning                                                                                         |
| <b>ML / Programming</b>    | PyTorch, JAX, TensorFlow, Scikit-learn                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Numerical Methods</b>   | Finite Element Methods (FEM; mixed Taylor–Hood elements); Meshless Methods (Peridynamics, Natural Element Method (NEM)); Delaunay/Voronoi discretization and Gauss quadrature; Newton–Raphson, L-BFGS and dynamic-relaxation solvers; automatic-differentiation residuals and tangents; matrix-free Newton–Krylov solvers and adjoint sensitivities; stability analysis (Legendre–Hadamard, second variation) |
| <b>Physics</b>             | Nonlinear elasticity; coupled multiphysics (diffusion–mechanics, electrodynamics, thermoelasticity); generalized continua (Cosserat, micropolar, higher-gradient); soft matter (liquid crystal elastomers, hydrogels, polymers)                                                                                                                                                                               |
| <b>Simulation Software</b> | FEniCSx, COMSOL Multiphysics                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Production / MLOps</b>  | End-to-end ML pipelines (data ingestion, training, checkpointing, evaluation); modular, standalone JAX/PyTorch packages with one-command, multi-seed sweeps; open-source library development; surrogate benchmarking against converged FEM ground truth                                                                                                                                                       |
| <b>Programming</b>         | Python (primary), MATLAB                                                                                                                                                                                                                                                                                                                                                                                      |

## RESEARCH & PROFESSIONAL EXPERIENCE

---

**Postdoctoral Associate — Yale University**  
*Department of Biomedical Engineering, New Haven, CT*

*Sept 2023 – Present*

- **Differentiable scientific computing:** Developed a differentiable, autodiff-based meshless peridynamics solver for nonlinear solid mechanics, modeling brittle fracture, pressure-driven (hydraulic) fracture, and delamination (Mode I/mixed-mode peel) without remeshing; achieved an **8 – 10 $\times$  runtime speedup** and **9.7  $\times 10^{-11}$  agreement** with a validated reference solution. Engineered a matrix-free dynamic-relaxation scheme to overcome Newton–Raphson tangent singularities and converge through unstable crack propagation. In collaboration with Prof. Ali Javili (Bilkent University); [project page](#).
- **Differentiable FEA (matrix-free Newton–Krylov):** Built an [open-source](#) PyTorch finite-element solver for large-strain hyperelasticity that never assembles the stiffness matrix: residuals and tangents come from automatic differentiation, and each Newton step is solved by preconditioned Krylov (conjugate-gradient) iteration using only Jacobian–vector products. Exact design gradients follow from the implicit function theorem with one adjoint solve. Matches FEniCSx to  $< 10^{-13}$  (relative); an element-wise operator variant makes the GPU tangent product **11.6 $\times$  faster** (40,000 Q4 elements, T4), and a complete 10,000-element solve takes **2.9 s** ([project page](#)).

- **Label-free physics-informed networks and neural operators for nonlinear solids:** Designed variational, energy-minimizing PINNs and DeepONets in PyTorch and JAX for liquid crystal elastomers (coupled finite-strain deformation and director fields), trained on the total potential energy over FEM/NEM discretizations (Delaunay/Voronoi, perforated domains) with Gauss quadrature and exactly enforced boundary conditions. A single operator trained over a load range predicts the full deformation and director fields in  $\sim 0.5\text{--}0.7$  ms,  $\sim \mathbf{35\text{--}250\times}$  **faster per query** than a warm FEM solve (156-node plane-strain benchmark, CPU; training repaid after  $\sim 100\text{--}800$  queries), with  $\sim 0.7\%$  (shear) and  $1.6\text{--}3\%$  (uniaxial) displacement error on unseen stretches against the same-mesh FEM reference. Derived quasiconvexity, rank-one convexity, and Legendre–Hadamard inequalities from the energy loss and used them as validation criteria: all stored trajectories are strongly elliptic strict local minimizers of the discretized energy ([project page](#)).
- **Data-free, discretization-consistent physics-constrained GNNs:** Trained graph neural networks to solve coupled multiphysics PDE systems as implicit time-steppers by minimizing the FEM weak-form residual directly, enforcing conservation laws and thermodynamic consistency with no labeled solution data. The graph is the mesh (or its Voronoi dual) and the network shares shape functions, quadrature and boundary conditions with the FEM solver, so each network is geometry-informed, and the weak-form residual serves as a built-in error monitor. Retrained per case. Demonstrated on two systems against reference Newton–Raphson FEM solutions, in both JAX and PyTorch (residuals parity-checked to  $\sim 10^{-12}$ ):
  - **PI-GCN for chemoelasticity (gel swelling):** Graph-convolutional network with residual connections for a transient, finite-strain, coupled 4-field system (displacement, pressure, chemical potential, concentration); block-weighted weak-form loss and Adam  $\rightarrow$  L-BFGS training, FEM and meshless NEM back ends, plate with a hole. Final-state error 0.24% in displacement and 0.73% in concentration over 50 implicit steps (441 nodes).
  - **PI-EPD-GNN for thermoelasticity:** MeshGraphNets-style Encode–Process–Decode network for transient heat conduction coupled to finite-deformation elasticity ( $\mathbf{u-p-\theta}$ ), with Taylor–Hood elements, a Voronoi message-passing graph, and a loss preconditioned by the rest-state tangent. Errors of 0.14% (displacement) and 0.011% (temperature) on a plate with a hole (PyTorch).
- **Inverse constitutive modeling / data-driven material discovery:** Developed PINN-based inverse frameworks in JAX and PyTorch for learning polyconvex stored-energy functions for isotropic and anisotropic materials, including fibrous composites and biological tissues, directly from FEA-generated datasets, with polyconvexity enforced as a hard architectural constraint.
- **PI-FNO — open-source neural operator deployment:** Designed, implemented, and open-sourced [PI-FNO](#), a Physics-Informed Fourier Neural Operator for 5-class valvular heart disease classification from phonocardiograms. Enforced physiological constraints (cardiac periodicity and S1/S2 frequency hierarchy) as hard architectural priors, with a learnable frequency mask initialized from cardiac acoustics and adapted end-to-end; achieved **97.5% test accuracy** across 1,000 recordings.
- **Multiphysics simulation infrastructure:** Built high-fidelity simulation frameworks in FEniCSx, COMSOL, JAX and PyTorch for fibrous biomaterials, hydrogel and elastomer swelling, thermoelasticity, and diffusion–mechanics coupling, with Newton–Raphson solvers using autodiff tangents. Implemented large-deformation swelling across multiple material models and compared penalty, Lagrange-multiplier, and augmented-Lagrangian constraint formulations. Currently extending the hydrogel swelling model to biological tissues and to delamination of layered hydrogel systems. Simulation datasets serve as ground truth for surrogate training and benchmarking.
- **Research leadership:** Led a multidisciplinary team of 4 researchers across Yale, Brown, and Stanford; mentored PhD and undergraduate students in PyTorch-based constitutive model discovery from biaxial experimental data.
- **Output:** 3 published · 4 under review · active peer reviewer.

## ENTREPRENEURIAL EXPERIENCE

**Co-Founder & ML Engineer — UC Berkeley (SkyDeck Pad-13 / CITRIS Foundry)**  
Berkeley, CA

*Jul 2022 – Sept 2023*

- Co-founded an AI startup for cardiovascular disease detection from phonocardiograms; accepted into two competitive Berkeley incubator programs (SkyDeck Pad-13 and CITRIS Foundry 2023).
- Developed, benchmarked, and delivered CNN + transfer-learning models for multi-class heart-sound classification; achieved **95% accuracy** on held-out test sets, meeting the startup’s clinical deployment target, and managed the full model lifecycle from data pipeline through evaluation on a startup timeline.
- Mentored an undergraduate student in building a hardware digital stethoscope paired with the deployed model.

## PH.D. RESEARCH — UNIVERSITY OF CALIFORNIA, BERKELEY

- **Graduate textbooks (2):** Co-authored *Principles of Continuum Mechanics* (World Scientific) and *Lecture Notes on the Theory of Plates and Shells. Classical and Modern Developments* (Springer).
- Derived quasicovexity, rank-one convexity, and Legendre–Hadamard inequalities for Cosserat elasticity, fibrous composites, and liquid crystals, establishing analytically rigorous output envelopes applicable as validation criteria for FEA and neural-network predictions.
- Developed constitutive models for fiber-reinforced composites and metamaterials capturing size-dependent Cosserat and strain-gradient behavior; implemented FEA in COMSOL and Python.
- Developed Kirchhoff-rod models for fibrous materials, fiber-reinforced thin-structure models for heart valves, and computational models of lipid bilayers reproducing biconcave red-blood-cell geometry.
- **Output:** 2 books · 21 peer-reviewed papers · 5 book chapters · Paul Naghdi Fellowship · Bob Steidel Fellowship · **Outstanding Graduate Student Instructor Award**.

## EARLIER RESEARCH EXPERIENCE

---

**Graduate Researcher — Pennsylvania State University**  
*Department of Engineering Science and Mechanics · State College, PA*

*Aug 2014 – May 2016*

- Experimental characterization of iron-based shape-memory alloys (DSC, tensile testing) and carbon-fiber/fiberglass composites; photoelastic stress analysis.

**Graduate Researcher — Isfahan University of Technology**  
*Department of Mechanical Engineering · Isfahan, Iran*

*Sept 2010 – Jan 2013*

- Proved thermodynamic inconsistency of the classical Brinson (1993) shape-memory-alloy model; derived a corrected constitutive framework validated against experimental data in MATLAB, Python, and Abaqus.
- Derived a thermodynamically consistent model for magnetic shape-memory alloys; single-calibration protocol reproduced 12 distinct magneto-mechanical loading cases. Predicted from thermodynamic reasoning that biaxial compression must produce hysteresis, then designed and executed experiments confirming it.
- **Output:** 8 peer-reviewed papers · **Best Thesis Award (across all degree levels: BSc, MSc, PhD)**.

## EDUCATION

---

**Ph.D., Mechanical Engineering — University of California, Berkeley** *2023*  
*Minors: Mathematics & Dynamics · 2 books · 21 papers · 5 chapters · Paul Naghdi Fellowship · Bob Steidel Fellowship*  
**Outstanding Graduate Student Instructor Award**

**M.Sc., Mechanical Engineering — Isfahan University of Technology** *2013*  
**Best Thesis Award (across all degree levels: BSc, MSc, PhD)** · 8 publications

**B.Sc., Mechanical Engineering — Isfahan University of Technology** *2009*

## CERTIFICATES & TRAINING

---

**Data Science Certificate — Flatiron School** *Nov 2021 – Aug 2022*  
*Machine Learning & Deep Learning · New York, NY (Online)*

- **Audio Emotion Detection:** CNN + transfer learning (EfficientNetB3/B7) for multi-class speech-emotion classification; mel-spectrogram preprocessing pipeline; **~99% test accuracy**.
- **Fake/Real News Detector:** NLP pipeline using NLTK, TF-IDF, GloVe, logistic regression, LightGBM, and TF-Hub transfer learning on 76,525 samples; **90% test accuracy, 0.90 weighted F1**.

## PUBLICATIONS & OUTPUT

---

- **Books (2):** *Principles of Continuum Mechanics* · *Lecture Notes on the Theory of Plates and Shells. Classical and Modern Developments*
- **33** peer-reviewed papers (32 journal · 1 conference) · 4 under review · 5 book chapters

- **Full list:** [Google Scholar](#)